BUILDING A USB RC METER

# Episode 3 Worksheet

## Auto-Ranging: Current Stages, Analog Switches and the ADC

*Work through these after watching the episode. Show your working; answers are on the last page. Values come from the 2014 thesis data unless stated.*

### Learning objectives checked

- Explain why a wide-range meter needs several current ranges
- Describe the 'sneak path' bug and how an analog multiplexer fixed it
- Convert an ADC count into ohms, and pick a sensible $V_{REF}$
- Explain ratiometric measurement and 2-wire vs 4-wire connections

### Questions

1. What voltage would Stage 1 (1.5 mA) need to measure 2 MΩ? What does this tell you?

2. With $V_{REF}$ = 3.8 V and a 10-bit ADC, what is one LSB? What would it be with $V_{REF}$ = 2.048 V?

3. Frame "3rdS400" arrives. Using 16.88 µA for Stage 3, compute R.

4. A 1.8 kΩ resistor is measured on Stage 1. How many ADC counts, and what's the rounding error in %?

5. Stage 1's current-setting resistance is 160 Ω + $R_{on}$. If $R_{on}$ rises from 345 Ω to 365 Ω, by what % does the current change?

6. Explain in one sentence why the design is ratiometric.

7. Your meter's leads plus socket add 4 Ω. What reading do you expect for a 10.0 Ω resistor, and how would a REL/zero function fix it?

8. Why could switching the PNPs OFF by pulling their bases to +5 V avoid the sneak path?

## Answers

1. V = 1.5 mA × 2 MΩ = 3000 V: impossible from 5 V, so smaller currents (more ranges) are required.
2. 3.8 / 1024 = 3.711 mV; 2.048 / 1024 = 2.0 mV.
3. 400 × 3.711 mV / 16.88 µA = 87.9 kΩ.
4. 1800 / 2.49 ≈ 723 counts; ±0.5 / 723 ≈ ±0.07 %.
5. Total 505 → 525 Ω, so the current falls by ≈3.8 %.
6. The measuring current and the ADC reference both scale with the same 5 V supply, so their ratio (the count) is almost independent of the supply.
7. ≈14 Ω. Shorting the probes stores ≈4 Ω, which is then subtracted from every reading.
8. With the base at 5 V and the collector ≤ 4.2 V, the base–collector junction is reverse-biased, so it can't conduct.

### Going further

**TI CD4051B datasheet ($R_{on}$, absolute max ratings):** https://www.ti.com/lit/ds/symlink/cd4051b.pdf

**Microchip PIC16F882/3/4/6/7 datasheet (ADC chapter):**
http://ww1.microchip.com/downloads/en/devicedoc/41291f.pdf

**Microchip Developer Help: ADC acquisition time and source impedance:**
https://developerhelp.microchip.com/xwiki/bin/view/products/data-converters/adc-specs/acquisition-time/

**All About Circuits: Kelvin (4-wire) resistance measurement:**
https://www.allaboutcircuits.com/textbook/direct-current/chpt-8/kelvin-resistance-measurement/

**Keysight: four-wire resistance measurement:** https://www.keysight.com/us/en/use-cases/measure-resistance-using-four-wire-measurement.html

**AVR Transistortester documentation (ratio method):**
https://www.mikrocontroller.net/articles/AVR_Transistortester